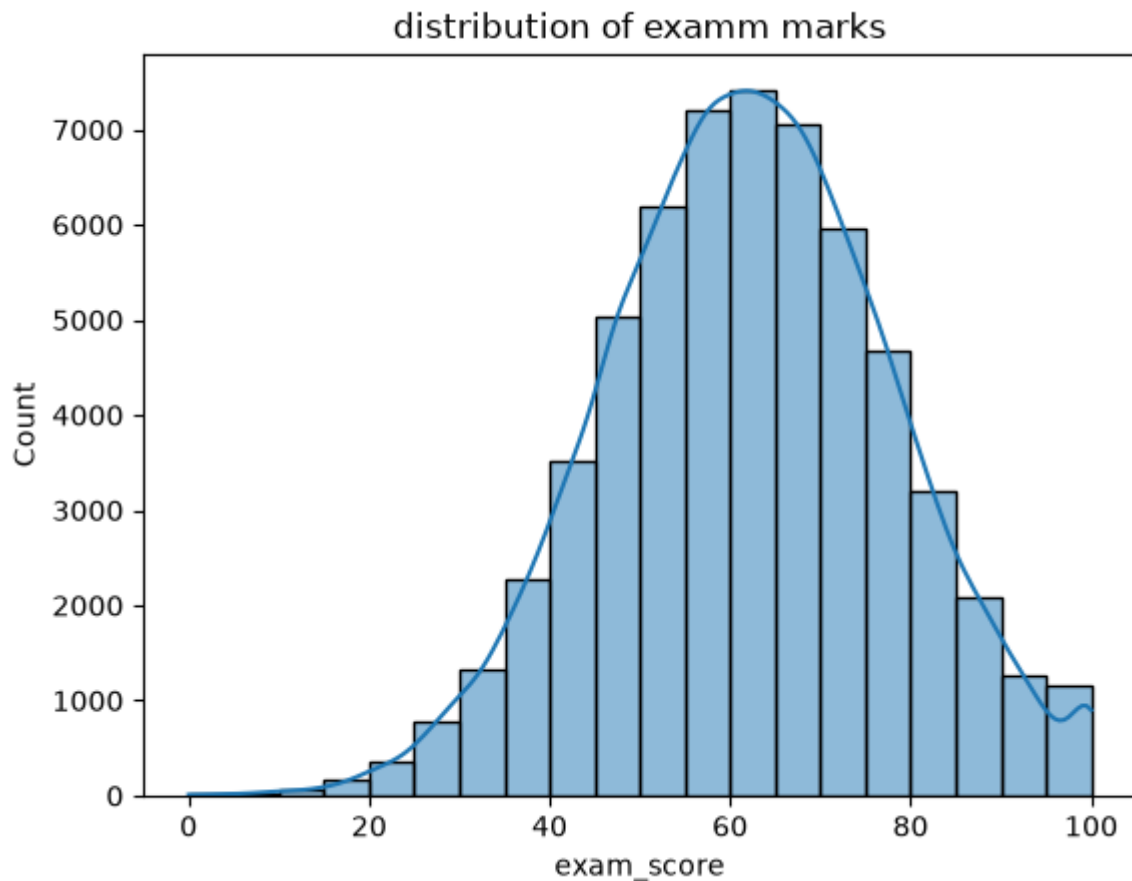


1. Distribution of marks



Observation

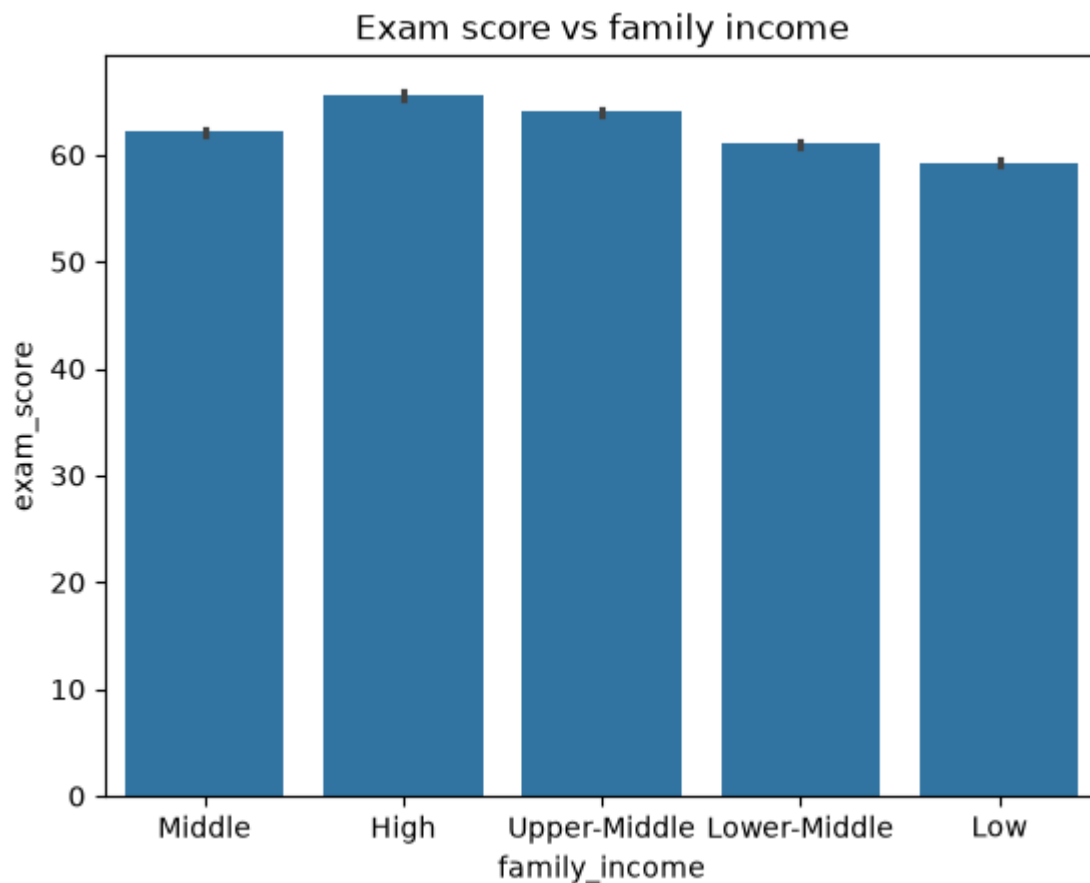
- The histogram shows a bell-shaped distribution with scores cantered around the 60 peak score.
- Student performance is mostly in the middle tier and extreme failure or high marks are rare.

Why it matters

- The histogram proves that the exam neither too difficult (demonstrated by high pass rate) or too easy(demonstrated by low frequency of high marks) for the students to understand.

Recommended action

2. family income vs Exam performance



Insight:

- The performance differences between students from low, middle, upper middle and high income bracket households are mostly negligible, suggesting income is not a big barrier.

Supporting evidence

- The bar chart shows that exam scores average around 65% for high income households, 59% and 62% respectively for low and middle income household students.

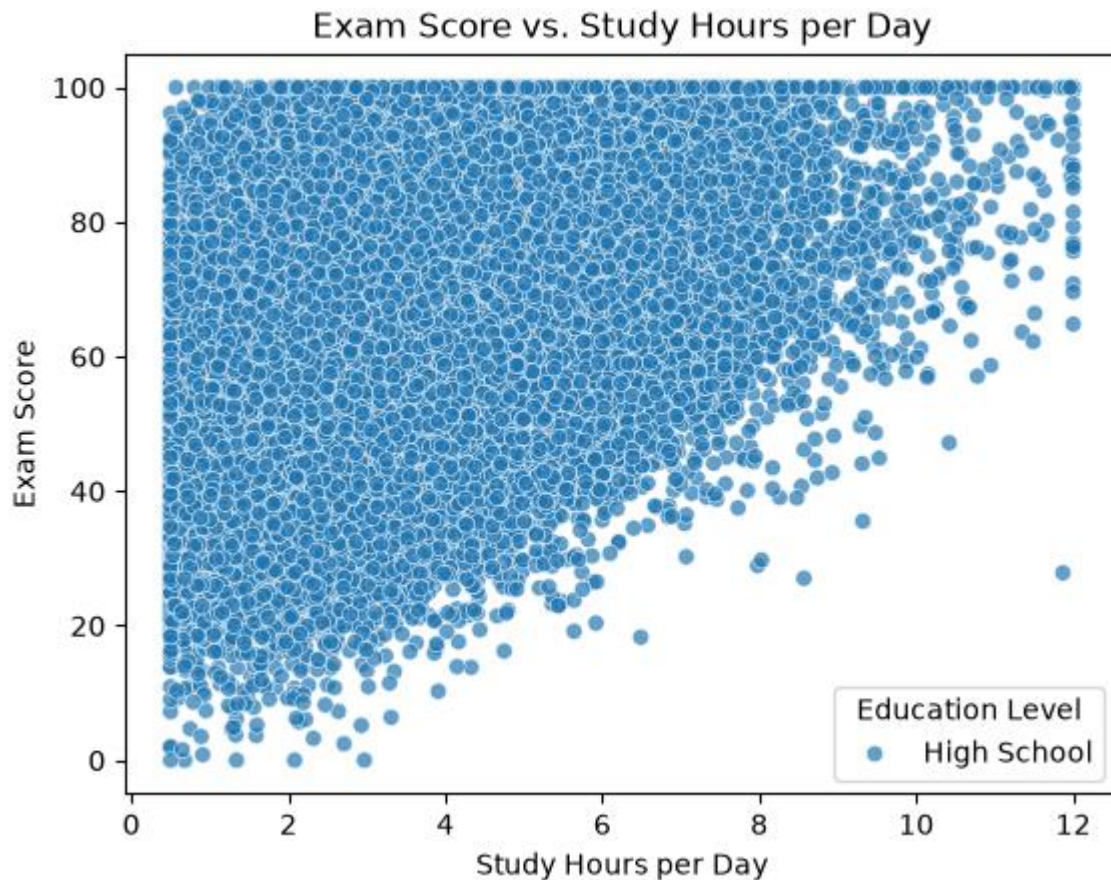
Why it matters

- Limited access to resources as a predictor of talent risks neglecting students from disadvantaged backgrounds.

Recommended action

- Ensure equitable resource allocation for all students, regardless of background.

3. Study hours and performance



Headline insight

- Daily study hours demonstrate a positive correlation with better exam outcomes.

Supporting evidence:

The scatter plot mapping study hours per day against exam scores shows that students studying more than 8 hours frequently achieve scores above 80, though lower study hours span a wide range of scores

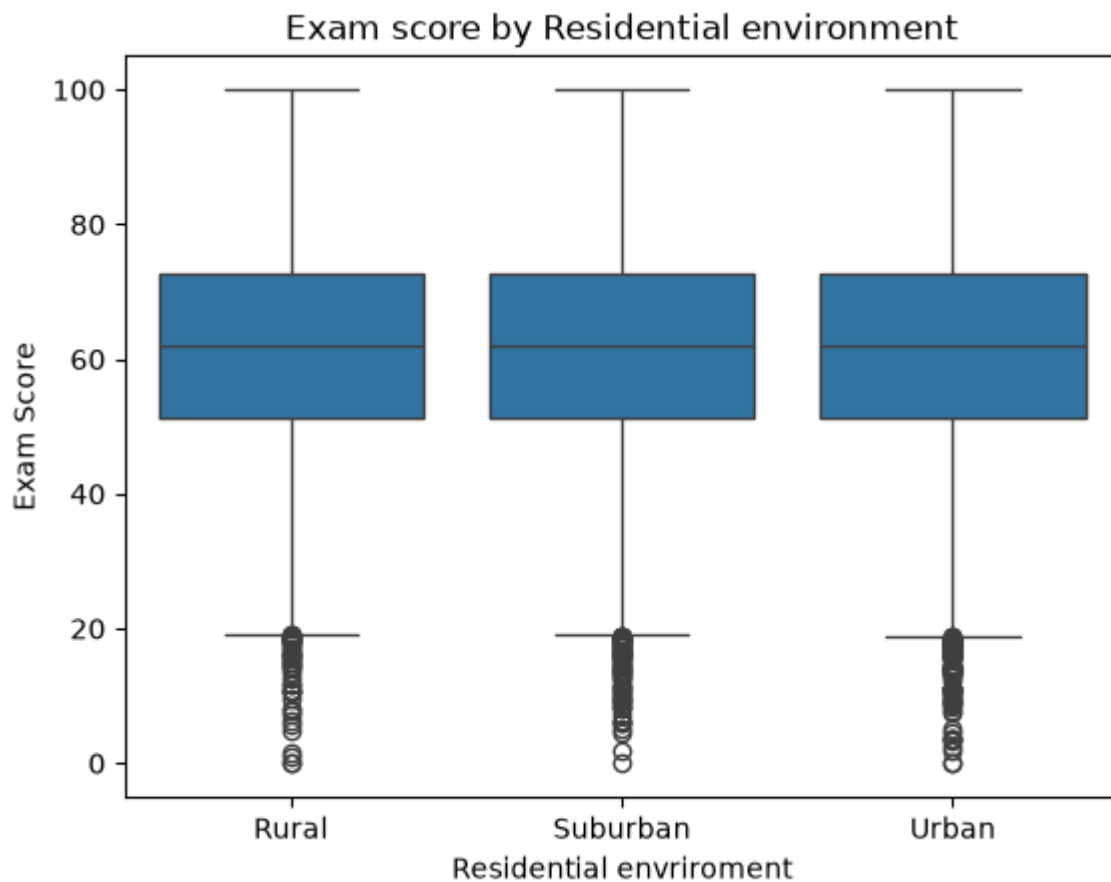
Why it matters

It demonstrates the relationship between studying and exam outcomes

Recommended

Coach students on effective study techniques and time-management strategies rather than simply pushing for longer hours at the desk.

4. Residential background vs exam outcomes



Insight

Residential environment has little systemic impact on overall exam score outcomes.

Evidence

The box plot comparing rural, suburban, and urban environments shows nearly identical median exam scores (~62) and interquartile ranges across all three groups.

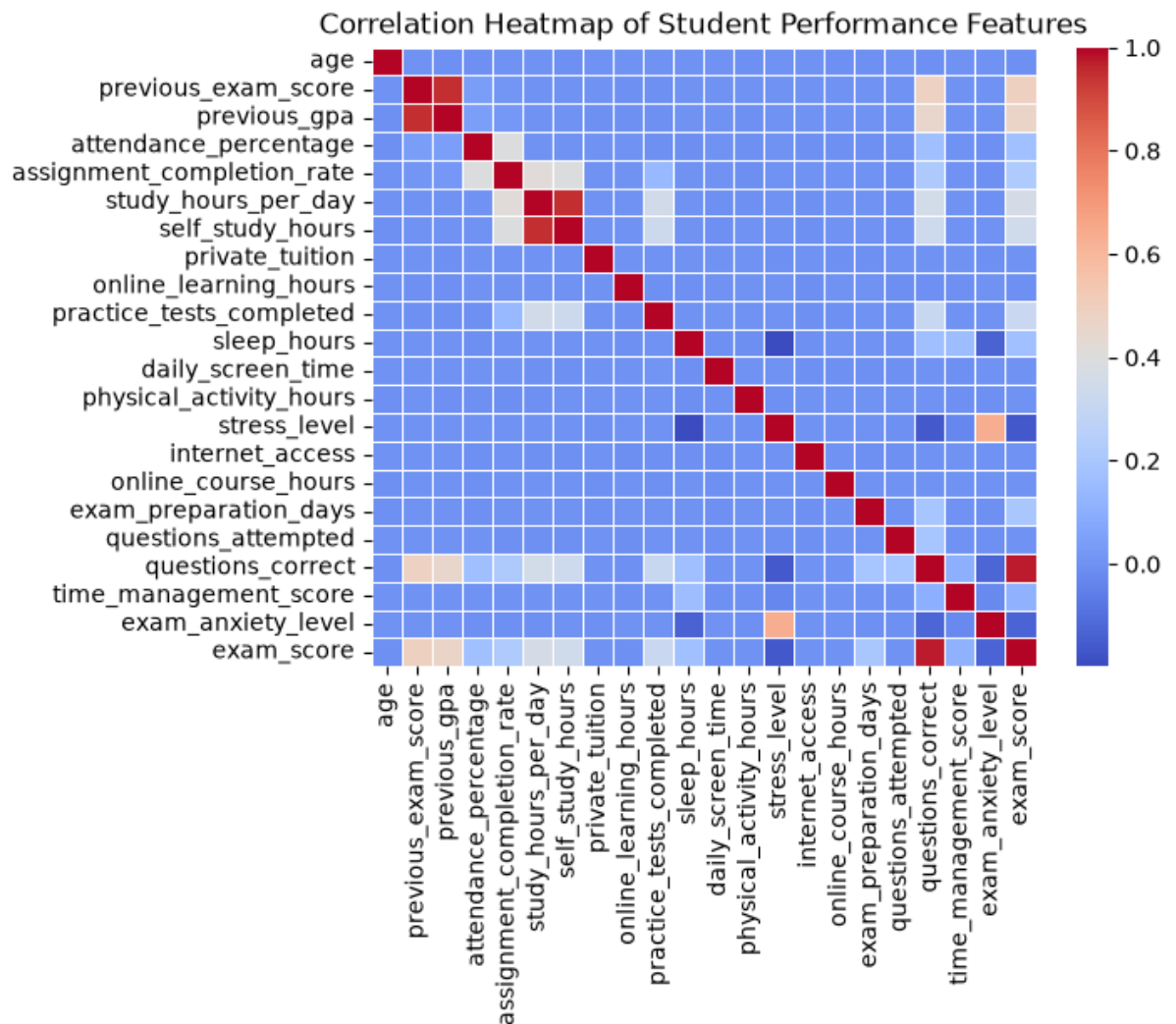
Why it matters

It demonstrates that residential background has no impact on exam outcomes.

Recommended action

Administrative interventions should focus primarily on individual students rather than broad community initiatives.

5. Core drivers of academic success



Headline Insight:

Historical academic markers and formative assessment metrics are the strongest predictors of current exam success.

Supporting Evidence:

The correlation heatmap highlights strong positive blocks connecting exam scores directly to previous exam scores, previous GPAs, and correct answers.

Why It Matters:

Recognizing these leading indicators allows educators to spot struggling students early through past performance data before final exams occur.

Recommended Action:

Build an early-warning dashboard tracking previous GPAs and formative test metrics to flag students who need proactive coaching.